



Windows 10's 'new' Firewall

The new Windows 10 comes with advanced features in its firewall. Read everything about it here: <http://dgit.in/winxadfirewall>



Don't miss this!

Check out here what happens when a bear eats up a GoPro here: <http://dgit.in/beareatcam>

lasers, however, experts remain sceptical from their initial description. Lasers inherently require line-of-sight and need to be focused, i.e., they can't transmit across a wide field of view.

Satellite constellations such as Iridium have existed long before the turn of the millennium. Satellite constellations are groups of artificial satellites that work in tandem to provide coverage all around the globe as evenly as possible. The orbits of the satellites are synchronised to maximise total coverage area and minimise interference. The satellites of Iridium were manufactured by Motorola



The Aquila Drone

and they provide telephony and communication via 66 active satellites in orbit. Iridium Communications sells access to their products and services to major companies. Another service is the Broadband Global Area Network, which is provided thanks to the INMARSAT constellation. Globalstar and Orbcomm are also constellations similar to Iridium. Coincidentally, Iridium, Globalstar and Orbcomm have all filed for bankruptcy at some point in their corporate existence, although they have still managed to survive. A seemingly completely altruistic venture is Outernet, which is a non-profit data broadcast service started in 2014. It transmits 200MB of open information every day to users, who are required to build their own receivers. However, this player appear to have stayed out of the Internet-for-All fray. A satellite constellation that does plan to reach out to the unconnected is O3b Technologies, which is backed by HSBC and Google. There is also a constellation named OneWeb, planned to launch by 2019, composed of a whopping 700 satellites!

Another means of getting Internet access to places where there is none using pre-existing infrastructure, is the TV

DID YOU KNOW? (STRATOSPHERE FACTS)

- Called so because it is vertically stratified (has air currents in layers)
- Has almost no turbulence, temperature rises as you move upwards
- Contains 90% of the ozone layer
- Contains bacteria, hence it's also part of the biosphere.

White Space spectrum. The shift to digital television means that many frequencies that were once licensed to analog TV broadcasters are now free and unused. Remember the time of cable when our televisions scanned across the whole spectrum provided (VHF and UHF) to obtain the list of channels? Television broadcasters occupied those bandwidths just as telecom companies occupy 3G/4G bandwidths. Moreover, they (analog broadcasters) had to use bands of frequencies separated by gaps in order to avoid interference, which showed up as the white noise between channels while scanning. These gaps are referred to as TV White Noise, and will be used (under certain guidelines) to broadcast the much higher throughput digital signals required for upto 15Mbps bandwidth Internet connections. Broadcasting over these frequencies (that are in the range of 54MHz to 806MHz) gives much better coverage (10km vs 100m radius) at a lower power cost compared to traditional Wi-Fi (2.4GHz and 5GHz), along with better penetration, earning this technology the nickname "Super Wi-Fi". Google and Microsoft have already invested in and

SATELLITE CONSTELLATIONS

- GPS - Global Positioning System (USA) - Navigation
- GLONASS - GLObal NAVigation Satellite System (Russia) - Navigation
- RapidEye (Germany) - Observation (Sensitive even to changes in chlorophyll content)
- INMARSAT (UK) - Telecommunications (Last to contact Malaysia Airlines Flight 370)

piloted projects that provide Internet access via the White Space spectrum in the United States of America, some places in the UK and in many countries in the African continent. Recently, Microsoft has partnered with the government of Andhra Pradesh to provide low cost Internet services to four schools in the Srikakulam district. Though our current crop of devices do not have the ability to directly access this spectrum of frequencies, that is all slated to change soon.

The Flat World

With so many electromagnetic waves flying around, the winner will be soon decided. But once the playing field is level, competition will undoubtedly increase, as will transparency and efficiency in almost every facet of life. Since currencies such as the dollar are based on words rather than actual resources, money isn't much different from circular logic. As often is the case, someone's gain is another's loss. Does that mean even in the Flat world, the rich will get richer and the poor will get poorer? Or will there be a natural requirement of a new world order? These questions are best left to the sociologists and anthropologists. We are here to discuss the technological aspects of a Flat world. One thing is clear, our infrastructure is expanding upwards. First to ascend was communication. Once it has proliferated to its fullest, will power be next?

With the latest advances in artificial neural networks, AI is improving in leaps and bounds. Perhaps unavoidably, it is being developed on the same networks that connect to the global Internet, which makes things pretty easy if ever an evil self-serving AI wanted to rule the world. On the bright side, since neural networks learn by example, the AI might respect us like we respect our parents. It might even invent a device that will allow digital technology to interface with our consciousness, though we will definitely invent such a device regardless. Maybe the social and technological aspects lead to the same answer: a collective human consciousness, moderated by technology? That is the ultimate equality among humans, when everyone can literally feel the same. First step: Internet for All!